

From Vine to Bottle

The annual growing season · The harvest · Every step from grape to glass · kazmicellars.com

Wine begins in the vineyard a full year before it reaches your glass. The vine follows a strict annual rhythm — dormancy, bud break, flowering, fruit set, véraison, harvest. Then the winery takes over: sorting, fermenting, aging, blending, bottling. Every step leaves a mark on the final wine. Understanding the process is understanding why winemakers make the choices they do.

PART ONE — THE ANNUAL GROWING SEASON

Dormancy

JAN–FEB

The vine is dormant — no leaves, no sap flow. Underground, roots store energy. Winemakers inspect canes and plan pruning strategy. Cold is essential: dormancy resets the vine's biological clock for the coming season.

Pruning

FEB–MAR

The most critical vineyard decision of the year. Pruning determines the vine's entire shape and potential yield. Every cut is deliberate — leaving too many buds = too many grapes, diluted quality. Too few = lost yield. Most canes from the previous year are removed, leaving only the selected fruiting canes or spurs.

Bud Break

MAR–APR

The first signs of life — tiny green buds push through the bark. One of the most anxious moments in the vineyard calendar. A single late frost at bud break can destroy an entire harvest. Producers light candles, use helicopters, and spray water to protect the fragile new growth from frost damage.

Shoot Growth

APR–MAY

Shoots grow rapidly — sometimes 2–3cm per day in warm conditions. The vineyard turns green quickly. Vignerons begin shoot positioning and tying, training new growth onto the trellis wires. The shoots will become next year's canes if selected at pruning.

Flowering

MAY–JUN

Tiny white flowers emerge. The vine self-pollinates — wind helps. Coulure (poor fruit set) and millerandage (uneven berry size) can occur if conditions are unfavorable during flowering. Rain, cold, or wind during this 10-day window can reduce the harvest significantly. A warm, calm flowering is every vigneron's prayer.

Fruit Set

JUN–JUL

Pollinated flowers swell into tiny green berries — hard, acidic, and full of tannin. The vineyard manager may do 'green harvest' (vendange verte) — dropping excess bunches to concentrate quality in the remaining fruit. What's removed now won't be in the bottle.

Véraison

JUL–AUG

The most visually dramatic moment in the vineyard. Red and black grape varieties change colour — from green to red, purple, or almost black — over 2–3 weeks. Berries soften, sugar accumulates rapidly, acidity begins to fall. This is the official start of the ripening sprint to harvest. White grapes don't change colour but soften and sweeten noticeably.

The final sprint. Sugar, acid, tannin, and phenolic ripeness are all monitored obsessively. The harvest date is one of the winemaker's most consequential decisions — too early = under-ripe; too late = over-ripe, high alcohol, low acid. Different varieties ripen at different times, so harvest can run 6–8 weeks across a large estate. Picking is done either by hand (gentle, selective) or machine (fast, economical).

PART TWO — HARVEST TO BOTTLE: THE WINEMAKING PROCESS

1 Receiving & Sorting

All wines

Grapes arrive at the winery and are inspected immediately. A sorting table (tapis de tri) allows workers to remove leaves, unripe clusters, damaged berries, insects, and anything that doesn't belong. Some estates do double sorting — once in the vineyard, once at the winery. At this stage, the quality ceiling for the wine is set. Nothing the winemaker does later can improve what arrives at the sorting table.

Key Decision: *Hand sorting vs machine sorting. Whole cluster vs destemmed. Speed matters — grapes oxidize quickly.*

2 Destemming & Crushing

Most wines

A destemmer-crusher removes stems from the grapes (optional — some producers use whole clusters). Crushing breaks the berry skin to release juice. Modern winemakers often prefer minimal crushing to protect the grape's integrity. Some use a gentle 'pigeage' (foot treading) for artisan production.

Key Decision: *Whole cluster fermentation (stems included) adds spice and structure — used by many top Pinot Noir and Syrah producers. Carbonic maceration (Beaujolais) uses whole uncrushed grapes fermented in CO₂.*

3 WHITE: Direct Press / RED: Cold Soak

Varies by colour

This is where red and white winemaking diverge sharply. **WHITE WINE:** Grapes are pressed immediately. The juice (must) is separated from the skins after a very brief (sometimes zero) skin contact. The fresher and cleaner the press, the more delicate the wine. **ROSÉ:** Brief skin contact of 2–24 hours gives the pink colour, then the juice is separated. **RED WINE:** The juice stays in contact with the skins. Some producers do a 'cold soak' (pre-fermentation maceration at low temperature, 5–10°C) for 1–5 days before fermentation begins. This extracts colour and soft fruity flavors without harsh tannins.

Key Decision: *Cold soak is debated — some producers swear by it for color and fruit depth; others skip it entirely.*

4 Primary Fermentation

All wines

Yeast converts grape sugar into alcohol and CO₂ — the transformation that turns juice into wine. Duration: 5–14 days for red wine; 2–4 weeks for white wine (often slower and cooler). Temperature control is critical — too hot and yeast dies or produces harsh flavors; too cold and fermentation stalls. **REDS:** The cap (solid grape skins floating on top) must be managed. Two methods: • Pump-over (remontage): pump juice from bottom, spray over top — gentle, extracts less tannin • Punch-down (pigeage): push the cap down by hand or tool — more extraction, fuller wines **WHITES:** Fermented in stainless steel (clean, fresh) or oak barrels (complexity, texture).

Key Decision: *Wild/native yeast vs commercial yeast: wild yeast ferments slower, adds complexity, carries risk. Commercial yeast is reliable and predictable. Natural wine producers exclusively use wild yeast.*

5 Pressing (Red Wine)

Red & Rosé

After fermentation, red wine is drained away from the skins ('free-run' wine) and the skins are then pressed to extract remaining liquid ('press wine'). Press wine is richer in tannin, color, and extract than free-run. The winemaker decides how much press wine to blend back in — more for structure, less for elegance. The remaining grape skins (marc/pomace) are discarded, composted, or distilled into grappa/marc brandy.

Key Decision: *Press wine decision is critical — too much makes the wine harsh; none leaves structure and concentration on the table.*

6 Malolactic Fermentation (MLF)

Most reds, some whites

A secondary bacterial fermentation that converts sharp malic acid (think green apple) into softer lactic acid (think cream). This occurs naturally after primary fermentation. Almost universal for red wines — it softens tannins and integrates structure. For whites, it's a deliberate stylistic choice: • Full MLF (oaked Chardonnay): butter, cream, round texture, lower acidity • Blocked MLF (Chablis, Sauvignon Blanc): crisper, more acidic, fresher • Partial MLF: a middle path — some producers use this for complexity with freshness.

Key Decision: *MLF is the single biggest winemaker decision affecting white wine style. Butter or no butter — this is where it's decided.*

7 Aging & Maturation

Most wines

Wine is transferred to its aging vessel — the choice shapes the final wine fundamentally: • New French oak (225L barrique): vanilla, toast, clove, smoke — strong flavor impact • Neutral oak (used 3+ times): structure without oak flavors — lets the wine speak • American oak: stronger vanilla, coconut, dill — Rioja's signature • Large foudres (500–3000L): micro-oxygenation without strong oak flavor • Stainless steel: pure fruit preservation, no oak, no oxygen — Sauvignon Blanc, Pinot Grigio • Concrete eggs: natural temperature regulation, micro-oxygenation, adds texture • Amphora/qvevri: ancient method — no oxygen, earthy minerality, used in Georgia and growing globally Duration ranges from 6 months (light whites) to 36+ months (Barolo, Brunello, Vintage Port).

Key Decision: *New oak vs neutral oak is one of the most consequential style decisions. Heavy new oak can overwhelm a delicate wine.*

8 Racking & Lees Management

Most wines

Racking moves wine from one vessel to another, leaving dead yeast cells (lees) and grape solids behind. Done periodically to clarify the wine and prevent off-flavors from decomposing lees. However, some lees contact ('sur lie' aging) is intentional: • Battonage (lees stirring): stirs lees back into wine — adds cream, body, yeasty complexity • Extended sur lie: common for Muscadet, Champagne, premium Chardonnay The 'angel's share' — evaporation through the barrel — concentrates the wine slightly. Barrels are topped up regularly to prevent oxidation.

Key Decision: *Lees aging is a quality decision. Extended contact adds complexity and texture. Racking too early removes potential.*

9 Fining & Filtration

Most commercial wines

Fining uses proteins (egg whites, casein, isinglass) or minerals (bentonite) to attract and precipitate unwanted particles, clarifying the wine. Traditionally: • Egg whites: classic for fine Bordeaux — gentle, clarifies without stripping • Bentonite: clay-based, removes unstable proteins from white wines • Isinglass: fish-derived — fining agent for whites and rosés Filtration physically removes particles through filter pads or membranes. Fine filtration ensures shelf stability but may strip texture. Many premium producers choose unfining and unfiltered wines, accepting some sediment in exchange for more complete, textured wine.

Key Decision: *Unfining/unfiltered is a quality statement in premium winemaking — but requires careful storage from buyer.*

10 Blending (Assemblage)

Most wines

If the wine is a blend — different varieties, different vineyard parcels, or different barrels — this is the winemaker's most creative act. Hundreds of combinations may be trialled before the final blend is set. The goal is a wine greater than the sum of its parts. For single-variety, single-vineyard wines, 'blending' means selecting the best barrels for the top wine and routing others to second wines or declassified bottlings. The second wine tradition (pioneered in Bordeaux) allows estates to maintain the quality bar of their grand vin by excluding any barrels that don't meet that standard.

Key Decision: *The assemblage decision is made in barrel, months before bottling. A great winemaker's nose and palate — irreplaceable.*

11 Bottling

All wines

The wine is transferred into bottles — an apparently simple step with significant risks. Cold stabilization (chilling to -4°C) may be done beforehand to precipitate tartrate crystals (harmless but alarming to consumers). Most wines receive a small addition of sulfur dioxide (SO₂) at bottling as a preservative and antioxidant. Closure choice matters: • Natural cork: traditional, allows micro-oxygenation, risk of cork taint (1–3%) • DIAM/technical cork: no taint risk, consistent • Screwcap: airtight, excellent for wines meant to drink young; stigma for fine wine • Glass stopper: premium, reusable, airtight Many estate wines require significant bottle aging before release — sometimes 1–5+ years.

Key Decision: *SO₂ management is the natural wine debate. Some producers add none — genuine risk; requires impeccable winemaking throughout.*

12 Bottle Aging & Release

Fine wines

Many wines continue to evolve dramatically in bottle before release and after. The estate holds the wine until they judge it ready — this can be months or years. Gran Reserva Rioja: minimum 5 years total. Brunello di Montalcino: minimum 5 years. Vintage Port: sometimes 20+ years before release by the quinta. In bottle, the wine undergoes 'reductive aging' — no oxygen, slow chemical transformation. Tannins soften and polymerize. Primary fruit fades. Tertiary aromas (leather, tobacco, truffle, earth, dried fruit) emerge. The wine becomes something fundamentally different from what was bottled. Great cellaring is part of the winemaking.

Key Decision: *The best bottles are made by both the producer and the buyer — the estate does its part, then trusts the wine to the cellar.*